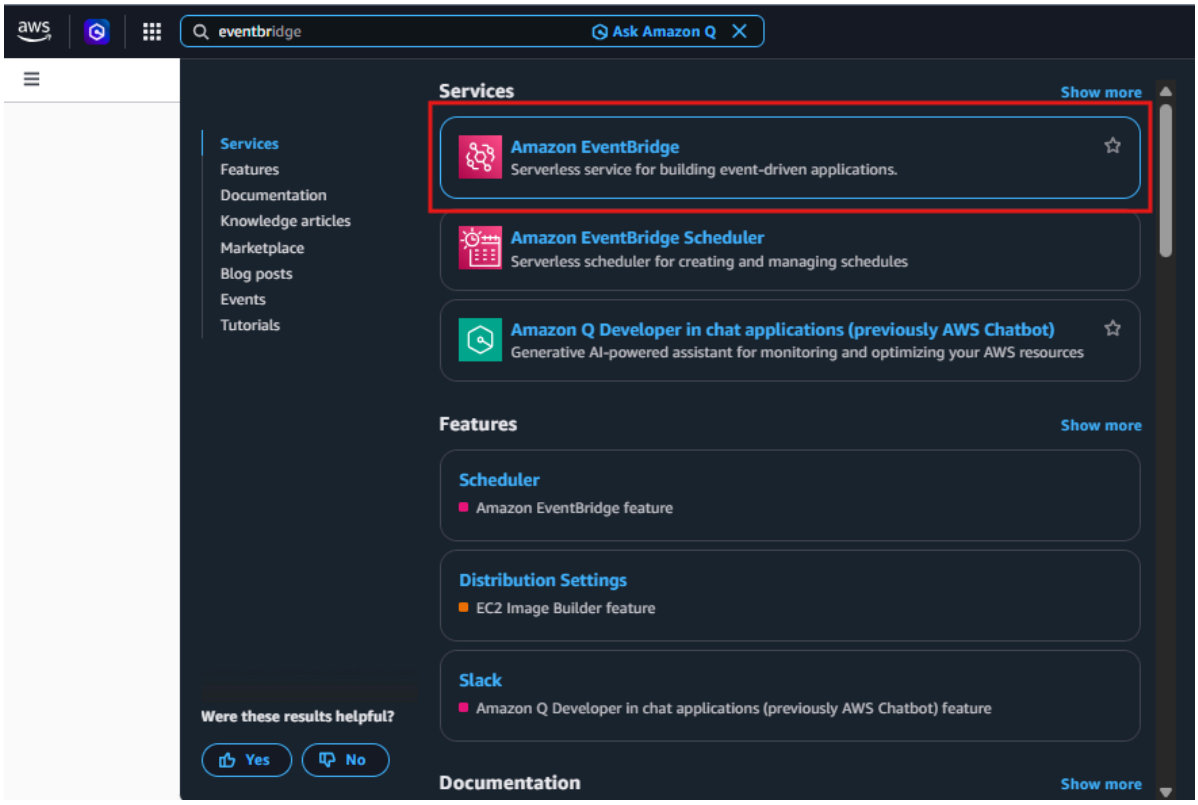
Lab: Schedule Step Functions Using Amazon EventBridge

Objective

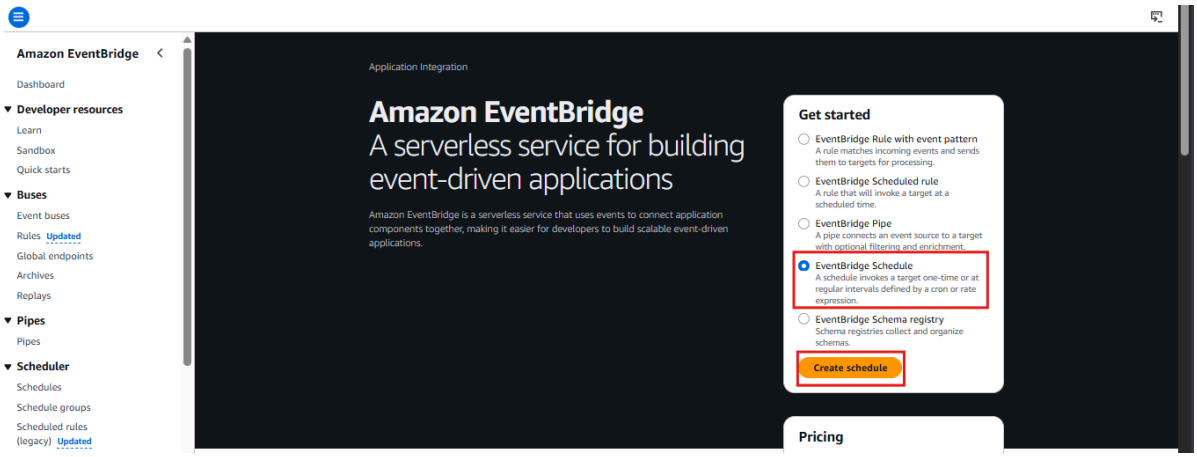
Schedule the execution of the Step Functions workflow using Amazon EventBridge so that the entire data pipeline runs automatically at fixed intervals.

Step 1: Open Amazon EventBridge

1. In the search bar, type **EventBridge**. —→ Select **Amazon EventBridge**.



2. Click on **EventBridge Schedule** —→ Click **Create schedule**.



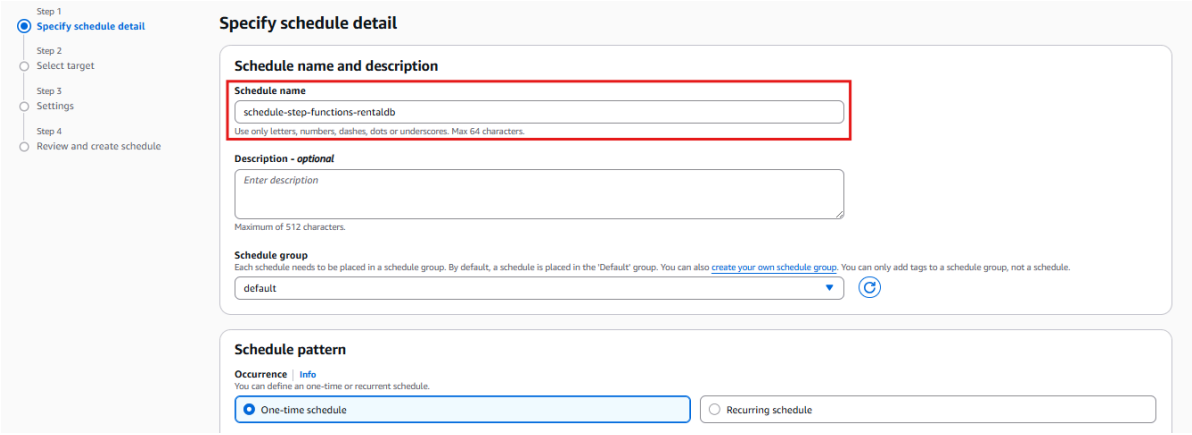
Step 2: Configure the Schedule

Schedule Name

1. Enter a meaningful name.

   Example:

   `schedule-step-functions-rentaldb`

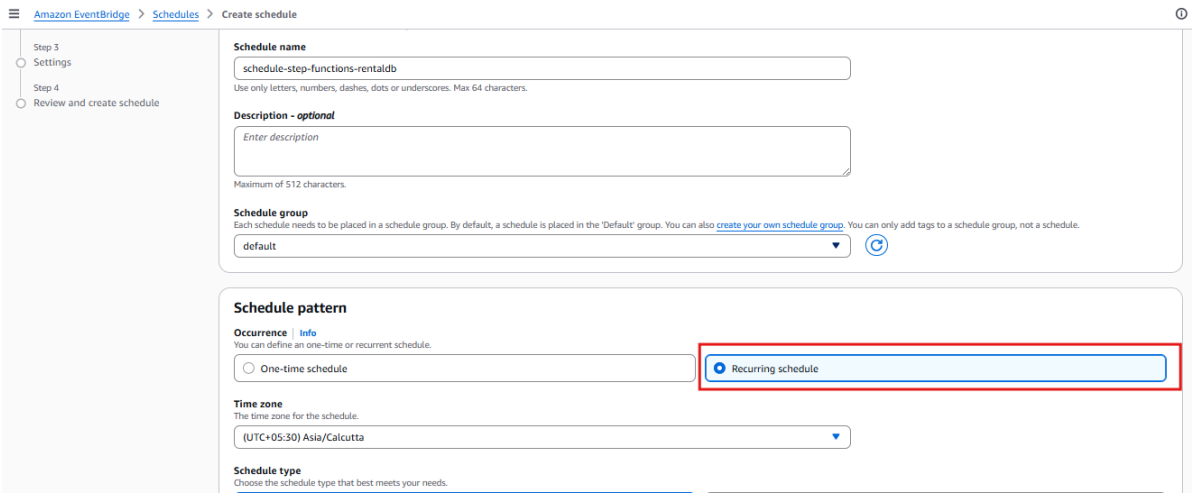


Schedule Pattern

You have two options:

- **One-time schedule**
- **Recurring schedule**

1. Select **Recurring schedule**.



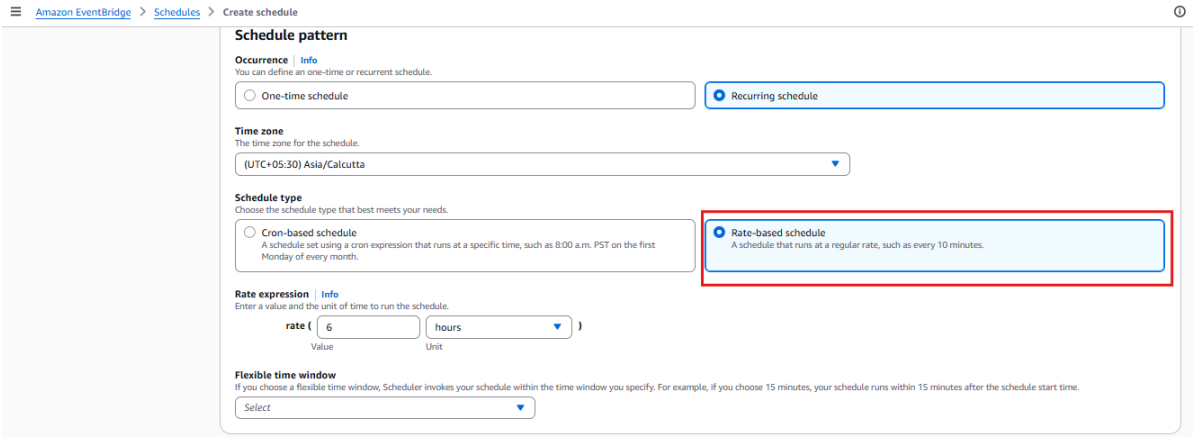
Schedule Type

You can choose between:

- **Cron-based schedule** (custom cron expression)
- **Rate-based schedule** (fixed time interval)

For this lab:

1. Select **Rate-based schedule**.
2. Set the interval to:
   - Value: `6`
   - Unit: `Hours`



This means the workflow will execute every 6 hours.

Flexible Time Window

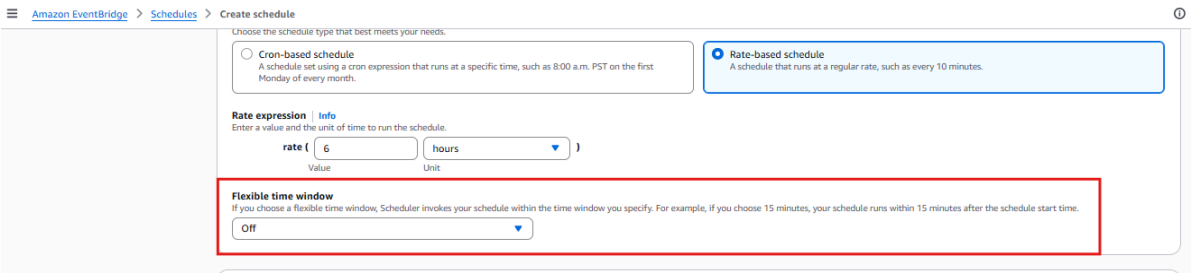
Flexible time window allows execution within a buffer period.

Example:

- If scheduled at 9:00 AM
- Flexible window = 5 minutes
- Execution may start anytime between 9:00 – 9:05 AM

For this lab:

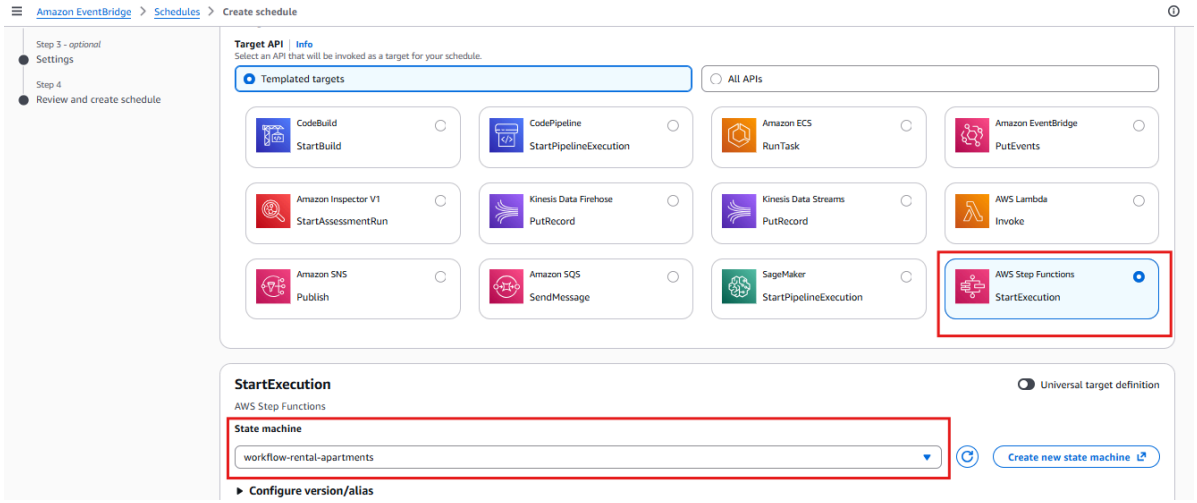
1. Turn **Flexible time window OFF**.
2. Click **Next**.



Step 3: Select Target Service

1. Under **Target type**, select **AWS service**.
2. Choose **Step Functions**.
3. From the dropdown, select the state machine created earlier:

   `workflow-rental-apartments`



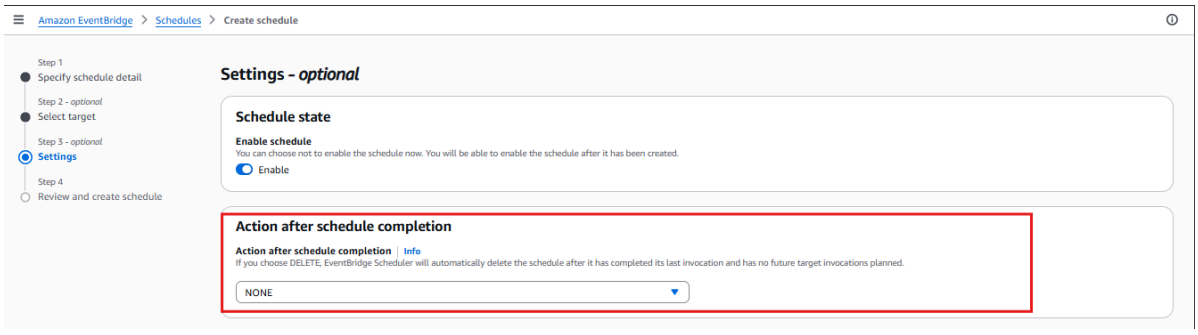
4. Click **Next**.

Step 4: Configure Additional Settings

Action After Completion

1. Select **None**

   (Since this is a recurring schedule, we do not want automatic deletion.)

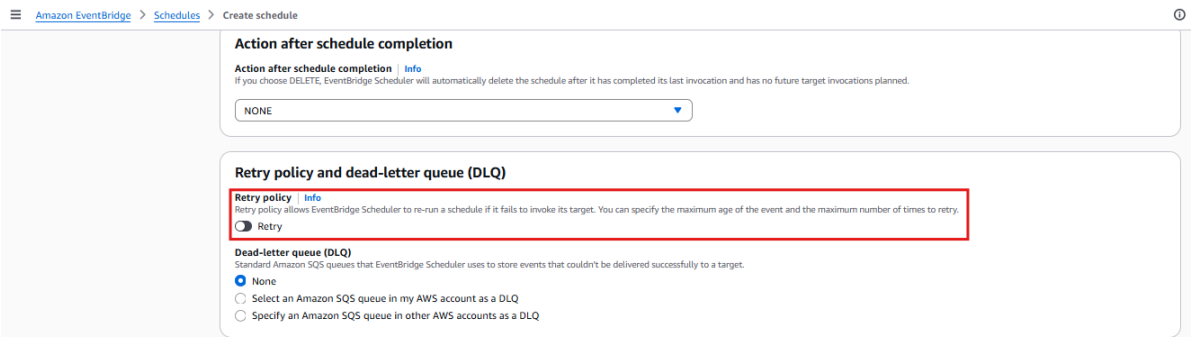


Retry Policy

EventBridge can retry execution in case of failure.

For this lab:

1. Disable Retry Policy (optional for learning environment).

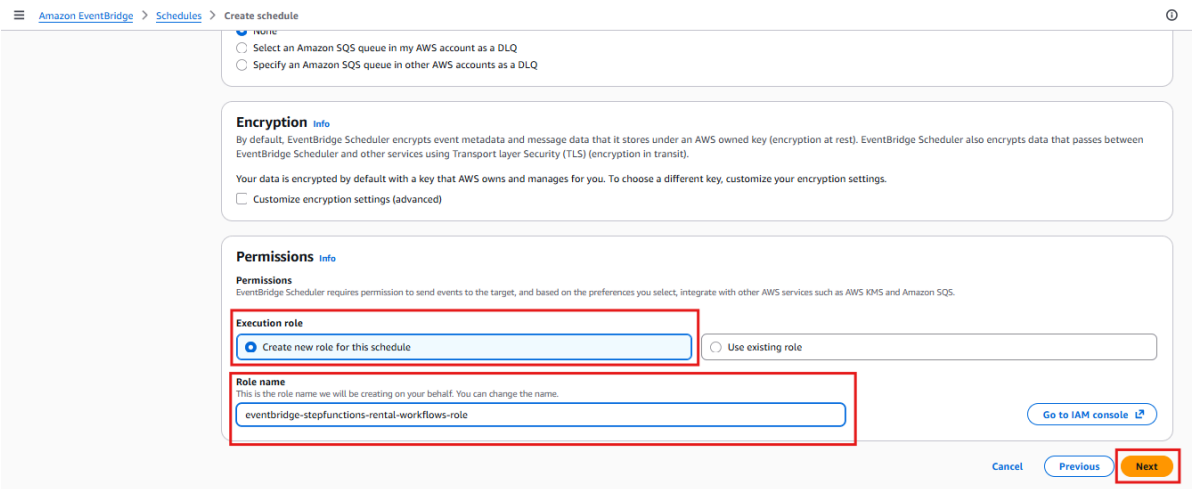


Execution Role

1. Select **Create new role**.
2. Provide a name for the role. Example:

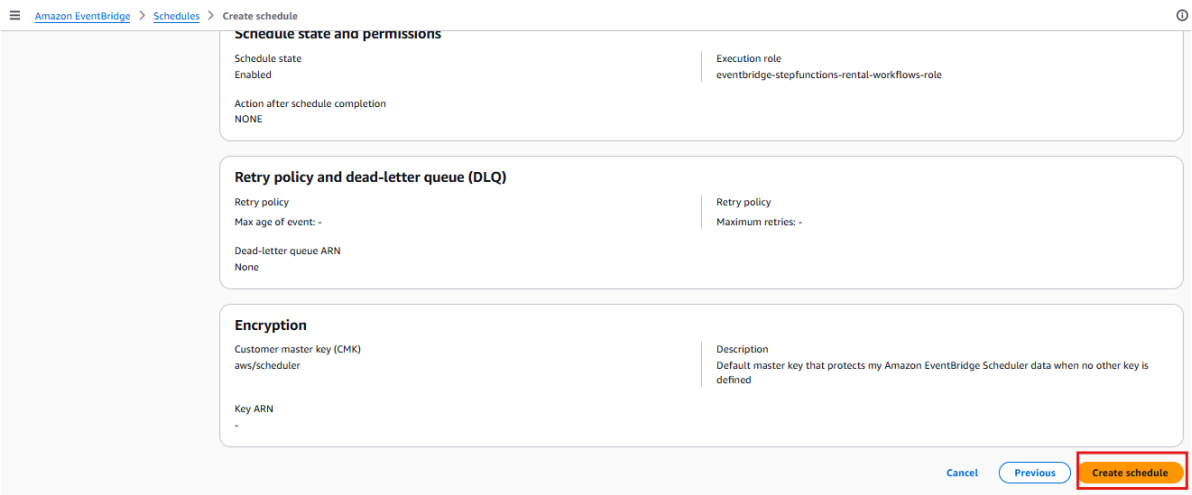
   `eventbridge-step-functions-rental-workflows-role`

   AWS automatically attaches the required permissions when created via console.
3. Click **Next**.



Step 5: Review and Create

1. Review the configuration.
2. Click **Create schedule**.



Verification

1. Navigate to **EventBridge** → **Schedules**.
2. Confirm the schedule status is **Enabled**.
3. Wait for the next execution window (6 hours).
4. Verify:
    - Step Functions execution starts automatically.
    - Glue jobs are triggered in sequence.
    - Workflow completes successfully.

Result

The Step Functions workflow is now automatically scheduled using Amazon EventBridge.

The complete batch pipeline (MySQL → Raw Layer → Process Layer) will execute every 6 hours without manual intervention.